



Lakshya JEE 2.0 (2024)

Definite Integration

DPP-02

- The value of $\int_0^4 |x-1| dx$ is:
 (1) $\frac{5}{4}$ (2) 5
 (3) 4 (4) 1
- $\int_0^{2\pi} (\sin x + |\sin x|) dx$ is equal to:
 (1) 0 (2) 4
 (3) 8 (4) 1
- If $f(x) = x - [x]$, for every real x , where $[x]$ is the greatest integer less than or equal to x . then, $\int_{-1}^1 f(x) dx$ is equal to:
 (1) 1 (2) 2
 (3) 3 (4) 0
- $\int_0^{\pi} \sqrt{1 + 4\sin^2 \frac{x}{2} - 4\sin \frac{x}{2}} dx$ is equal to:
 (1) $\pi - 4$ (2) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$
 (3) $4\sqrt{3} - 4$ (4) $4\sqrt{3} - 4 - \frac{\pi}{3}$
- The value of $\int_0^{\sqrt{2}} [x^2] dx$, where $[.]$ is the greatest integer function, is:
 (1) $2 - \sqrt{2}$ (2) $2 + \sqrt{2}$
 (3) $\sqrt{2} - 1$ (4) $\sqrt{2} - 2$
- If $f(x) = \begin{cases} 2x^2 + 1, & x \leq 1 \\ 4x^3 - 1, & x > 1 \end{cases}$, then $\int_0^2 f(x) dx$ is equal to:
 (1) $47/3$ (2) $50/3$
 (3) $1/3$ (4) $47/2$
- $\int_{-a}^a x|x| dx =$
 (1) $\frac{a}{3}$ (2) $\frac{a^2}{3}$
 (3) $\frac{a^2}{2}$ (4) 0
- The value of $\int_{-\pi/2}^{\pi/2} \log\left(\frac{2 - \sin\theta}{2 + \sin\theta}\right) d\theta$ is
 (1) 0 (2) 1
 (3) 2 (4) None of these
- The value of $\int_{-\pi/4}^{\pi/4} \log(\sec\theta - \tan\theta) d\theta$ is:
 (1) 0 (2) $\frac{\pi}{4}$
 (3) π (4) $\frac{\pi}{2}$
- The value of $\int_{-\pi/2}^{\pi/2} (x^3 + x \cos x + \tan^5 x + 1) dx$ is
 (1) 0 (2) 2
 (3) π (4) None of these
- If $\int_{-a}^a (\alpha \sin^5 x + \beta \tan^3 x + \gamma \cos x) dx$, where α, β, γ are constants, then the value depends on
 (1) γ, a (2) α, β, γ, a
 (3) α, β, a (4) β
- $\int_{\ln \lambda}^{\ln(\frac{1}{\lambda})} f\left(\frac{x^2}{3}\right) (f(x) + f(-x)) \frac{dx}{g(3x^2)(g(x) - g(-x))} = \dots\dots\dots$
 Where $\lambda > 1$.
 (1) 0 (2) 1
 (3) λ (4) $1/\lambda$

13. $\int_{\pi/6}^{\pi/3} \tan x \, dx + \int_{1/\sqrt{3}}^1 \tan^{-1} x \, dx = a - b + \ln \sqrt{2}$

then a/b is

- | | |
|--------------------------|---------------------------|
| (1) $\frac{\sqrt{3}}{2}$ | (2) $\frac{3}{2}$ |
| (3) $3\sqrt{3}$ | (4) $\frac{3\sqrt{3}}{2}$ |

14. The value of the integral $\int_{-1}^1 \frac{x^7 - 3x^5 + 7x^3 - x}{\cos^2 x} \, dx$

is:

- | | |
|---------------------|-------|
| (1) $\frac{\pi}{2}$ | (2) 0 |
| (3) $\frac{\pi}{4}$ | (4) 2 |

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (2)
2. (2)
3. (1)
4. (4)
5. (3)
6. (1)
7. (4)

8. (1)
9. (1)
10. (3)
11. (1)
12. (1)
13. (4)
14. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>